

Bharat Light & Power: Optimizing industrial equipment productivity and improving asset health with Google Cloud

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About Bharat Light & Power



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With Google Cloud, Bharat Light & Power is achieving its objective of becoming one of the best companies in the world at using enterprise AI and

Bharat Light & Power Pvt Ltd (BLP) is one of the leading renewable energy generation and technology companies in India. It builds, owns, and operates utility-scale renewable energy power plants to generate clean energy through wind and solar; provides distributed solar solutions to large commercial and industrial consumers in India, and, through its AI-based IoT platform, Orion, delivers monitoring, reporting, and predictive intelligence solutions across industries all over the globe. Through its control

Internet of Things
improve producti
reliability, quality,
sustainability acr
range of industrie
sectors, including
automotive, ener
manufacturing, p
supply chains,
infrastructure, an

center in Bangalore, Bharat Light & Power
leverages big data, the cloud, mobility, and
advanced analytics to help customers build
smarter operations.

Industries: Technology

Location: India

Google Cloud results

- Supports monitoring, reporting, and analytics for 2,000 wind turbines generating energy across several countries
- Gains ability to increase power generation,

Proc
from
data
578 turbines
and 200,000
data points in
4,000 sensors
per day

Pub/Sub (<https://cloud.google.com/pubsub/>),
Google Cloud (<https://cloud.google.com/>) flow/),
Cloud Functions
Cloud IoT Core (<https://cloud.google.com/iot-core/>)
Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)
Cloud Bigtable
Dataflow (<https://cloud.google.com/dataflow/>)
TPU (<https://cloud.google.com/edge-tpu/>)
Cloud Functions
(<https://cloud.google.com/functions/>)
BigQuery (<https://cloud.google.com/bigquery/>)
I tell us your challenge. We're here to help.
Cloud Bigtable (<https://cloud.google.com/bigtable/>)
Edge TPU (<https://cloud.google.com/edge-tpu/>)

**predict
component
failures,
calculate
remaining
useful life of
components,
and reduce
maintenance
costs**

- **Increases
manufacturing
productivity by
at least 5%
and reduces
costs by 10%**

As community and business concern over global warming, sustainability, and energy security continues to rise, companies such as Bharat Light & Power (BLP) (<https://www.blp.co.in/>) are working on answers. Founded in 2010 and located in Bangalore and Delhi, BLP is one of the leading clean energy generation companies in India. According to Tejpreet Singh Chopra, Chief Executive Officer of BLP, the business started with the aim of delivering clean energy from renewable sources such as wind, solar, biomass, and hydro to the 300 million residents of India that did not have reliable access to power at that time.

However, BLP soon began experiencing problems ensuring availability of the wind turbines that powered utility-scale wind power generation across

the country. “While we were investing large amounts building wind farms, the failure of the smallest component could shut down an entire wind turbine and compromise our efficient operation,” says Chopra. “A \$5,000 battery could fail and shut down a \$2.5 million piece of equipment!”

BLP transformed its legacy thinking about man, machine, technology, and process, and implemented an AI and Internet of Things (IoT) project. The business used hundreds of tags from its wind turbines to capture and send machine behavior and performance data to a control center. This data triggered an 18-month project to create AI and machine learning algorithms that would enable engineers to predict component failures in wind turbines, improve generation, and adjust maintenance and replacement schedules accordingly.

This activity provided the foundation of a technology business—one of the three companies that comprise BLP—that delivers enterprise AI for industrial uses over a platform, branded Orion. “We provide an end-to-end solution that enables businesses in industries such as transportation, logistics, ports, infrastructure, supply chain manufacturing, steel, and automotive to improve productivity,” says Chopra.

As the business expanded, it began to extend beyond its core “AI for industry” mission. It recruited an IoT team to help factory owners and operators enable the programmable logic controllers (PLC) and supervisory control and data acquisition

(SCADA) systems that run disparate equipment and machines—such as production line machines of different ages and countries of origin—to talk to each other and provide usable data and insights.

Agility and adaptability key

BLP's management team realized early that the business needed to be agile and adaptable to keep pace with changes in technology. "We knew we would have to effectively destroy and remake the business every 18–24 months to remain relevant, so we needed a service that could support our dynamic infrastructure, data, and AI needs," says Chopra.

The business started operations on a cloud service but quickly ran into problems. "We found in the world of industry—the vast amounts of data, the variety of sources of data, and the complexity of insights required—created a very different set of challenges relative to the consumer technology environment," says Chopra. "Our cloud provider could not provide an architecture that made sense for an industry-focused solution."

BLP's Domain Expertise lies in.....



IOT & Data Collection



Smart Analytics



Energy Efficiency



Rotating Machines
Excellence



Resource/ Asset Tracking
& Tracing



Video Analytics & Image
Processing



Electrical Drives –
Predictive Intelligence



Smart Buildings

Google Cloud to power AI and visual analytics

Within two years, BLP advanced its strategy and focused on using open source to reduce the cost of its architecture. As Orion matured and its take-up grew, the business began looking at multinational cloud services to run the forthcoming version 3.0 of the platform.

BLP found Google Cloud provided the best fit for its needs for a range of reasons, including the ease of use of Google Cloud services—likened by the BLP technology team as “the equivalent of a consumer app experience”—and its high-quality database services, high availability, fast response times, and the power to run the platform’s AI and visual analytics.

BLP established as its key objective a 20% reduction in costs over its previous cloud service and availability levels exceeding 99.9%. The business completed the deployment in June 2019 following discussions and input from Google Cloud’s engineering and architecture experts prior to and during the first stage of the project.

IoT Core (<https://cloud.google.com/iot-core/>)—a managed service that allows organizations to connect, manage, and ingest data from dispersed devices—is the cornerstone of the architecture supporting the latest latest version of the Orion platform, Orion 4.0. BLP has also created a data pipeline based on Pub/Sub (<https://cloud.google.com/pubsub/>) event ingestion and

delivery and Dataflow

(<https://cloud.google.com/dataflow/>) data processing.

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

provides a high-performance NoSQL database service for the platform's analytical workloads and

BigQuery (<https://cloud.google.com/bigquery/>) delivers a powerful analytics data warehouse. Cloud Functions

(<https://cloud.google.com/functions/>) enables the business's developers to build event-driven serverless applications.

This architecture currently captures, processes, analyzes, and reports on 8 million data points in 578 turbines around the world per day and processes data from 200,000 data points in 4,000 sensors per day.

The Google Cloud architecture supports the data visualization and reporting and the AI and machine learning-powered products created by BLP. These reports and products enable users to monitor remote assets and use AI-based analytics to predict machine failures before they occur, sequence failure events, and maximize equipment uptime.

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*—Tejpreet Singh Chopra,
Chief Executive Officer,
Bharat Light & Power*

A range of benefits to customers

With Google Cloud, BLP is delivering projects with a range of benefits to customers. “We’ve enabled one of the largest electrical companies in India to compare production line performance by capturing data from PLC and SCADA systems and extracting it to Google Cloud for processing, analysis, and reporting,” says Chopra. “For another customer, we’ve deployed an IoT system that has saved them about \$200,000 in energy costs over six months.”

The business is also providing monitoring, reporting, and analysis to predict likely failures of gearboxes, bearings, generators, and blades in 2,000 wind

turbines—that provide close over 2 GW of wind power—in countries such as France, Germany, Italy, India, Portugal, Spain, the United Kingdom, and the United States.

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*—Tejpreet Singh Chopra,
Chief Executive Officer,
Bharat Light & Power*

Furthermore, BLP is providing visual analytics to help one of the largest ports in the world detect when workers are not wearing helmets or safety equipment. “We also do a lot of inventory track-and-trace work to help companies improve supply chains and help factories keep track of tooling through Bluetooth low-energy technologies,” says Chopra.

With Google Cloud, BLP is now ideally placed to execute its business strategy of helping customers improve productivity and increase growth, control and reduce costs, and enhance quality and safety. “We are realizing this strategy by working to become the best company in the world at using AI to predict machine failure,” says Chopra. “Our underlying technology strategy entails enabling the most advanced IoT hardware used by industry to talk to software and the cloud, using the most powerful AI cloud around—which is why we chose to marry our AI algorithms with Google Cloud—and delivering insights through visualization.”

“The next wave
of

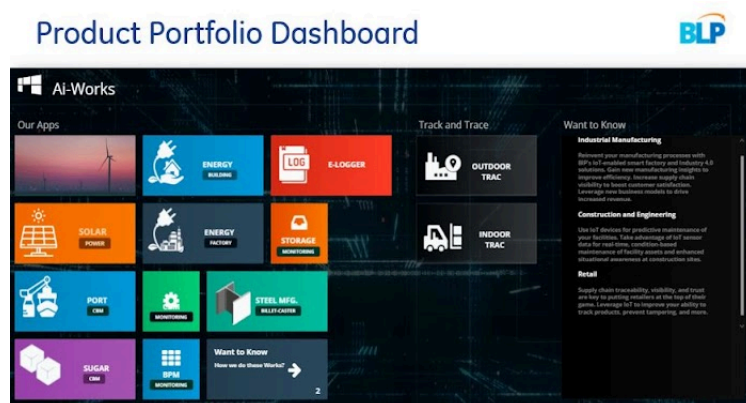
manufacturing
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world."

—*Tejpreet Singh Chopra,*
Chief Executive Officer,
Bharat Light & Power

Testing Edge TPU

The business is now testing Edge TPU (<https://cloud.google.com/edge-tpu/>) to run AI at the edge in high-performance, small-footprint, and low-power ASIC environments. "We think that will be the next big revolution as TPU costs come down," says Chopra. "It will be a considerable benefit performing AI at the edge rather than involving the full infrastructure of the cloud."

BLP is keen to build on its existing relationship with and use of Google Cloud to further transform manufacturing worldwide. “Manufacturers are pushing the boundaries of quality and cost through initiatives such as Lean Six Sigma, Zero Defects, poka-yoke, just in time, and other methodologies and approaches,” says Chopra. “The next wave of manufacturing improvements are coming through industry 4.0, AI, IoT, and Google Cloud’s analytics, databases and other services. These are ideal to power this change in the industrial world.”



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our team to
see how
Google Cloud
can help your

business.